



2024 Year in Review

Internet patterns and trends observed through Cloudflare's global network during 2024

17% Global Traffic Growth and Starlink's 3.3x Surge



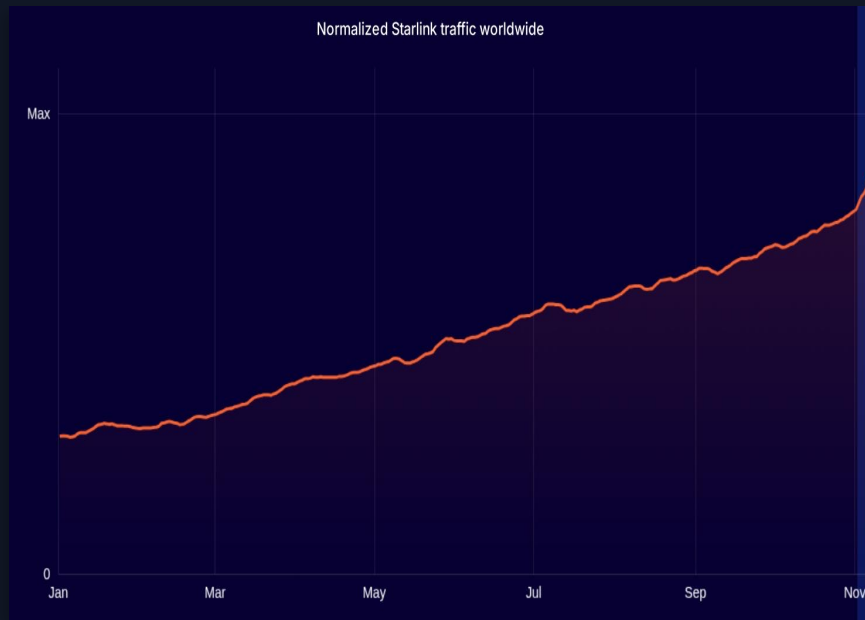
17%

Year-over-Year Global Internet Traffic Growth



3.3x

SpaceX Starlink Traffic Surge in 2024



Trend line starts mid-January to normalize for post-holiday activity. Starlink trend scaled relative to peak observed request volume.

Google Leads Top 10 Services as Generative AI Rises

RANK	SERVICE
#1	Google
#2	Facebook
#3	Apple
#4	TikTok
#5	AWS
#6	Microsoft
#7	Instagram
#8	YouTube
#9	Amazon
#10	WhatsApp



Google #1 for 2nd consecutive year

Rankings based on aggregate 1.1.1.1 DNS resolver data across all categories

Rank	Service
#1	Google
#2	Facebook
#3	Apple
#4	TikTok
#5	AWS
#6	Microsoft
#7	Instagram
#8	YouTube
#9	Amazon
#10	WhatsApp



Generative AI continued its rapid rise throughout 2024, while the Metaverse receded into the background.

Post-Quantum Encryption Jumps from 2% to 13% of TLS

1.3



33% iOS traffic **89%** Google search **Chrome** #1 browser



2% → 13%

Post-quantum encryption of TLS 1.3 traffic after Chrome 124 default enablement (Apr 2024)



21%

HTTP/3 adoption — faster connections, packet-loss mitigation, encryption by default via QUIC



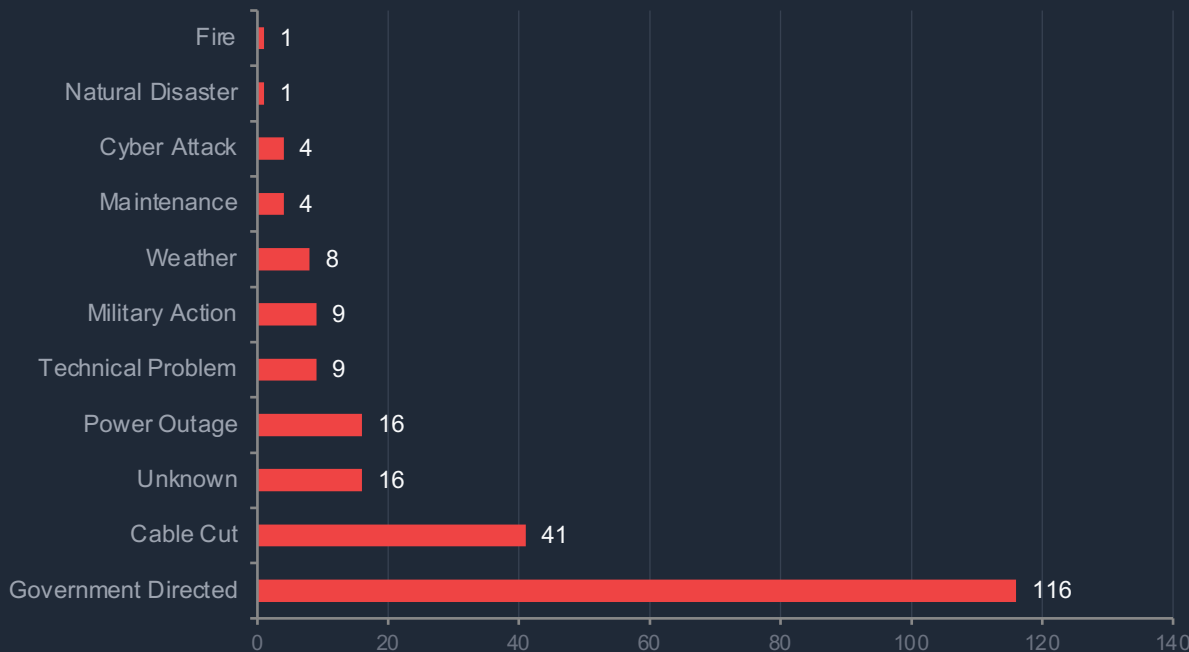
Go — #1 API Client

~12% of automated API traffic. Node.js, Python (9.6%), Java (7.4%), .NET (3.6%)

225 Major Disruptions: Government Shutdowns Lead Outage Causes

225

Major Disruptions
Observed in 2024



Government-directed shutdowns (116+) accounted for more than half of all disruptions. Cable cuts (41) and power outages (16) follow.

6.5% of Traffic Mitigated: Persistent Threats and Bot Concentration

 **6.5%**

Traffic
Mitigated

 **#1**

Gambling/Games
Most Attacked

 **68%**

Bot Traffic from
Top 10 Countries

 **4.3%**

Emails Found
Malicious

Log4j (CVE-2021-44228)

First disclosed December 2021, this critical vulnerability continues to be actively targeted by attackers more than two years later — underscoring the persistent nature of widely known vulnerabilities.

Email Security Threats

43% of malicious emails contained deceptive links
.bar TLD originated the largest share of malicious & spam email



RPKI valid routes increased by **6.4%** — advancing Internet routing security.

Five Takeaways Shaping the 2025 Internet Outlook

01



Sustained Traffic Growth

Global Internet traffic grew 17% YoY, with Starlink surging 3.3x as satellite connectivity expands to underserved regions.

02



Post-Quantum Momentum

Adoption leapt from ~2% to 13% of TLS 1.3 traffic following Chrome 124's default enablement in April 2024.

03



Connectivity Fragility

Government-directed shutdowns led 225 major disruptions, followed by cable cuts (41) and power outages (16).

04



Persistent Security Threats

6.5% of all traffic was mitigated. Gambling/Games was the most targeted sector, and Log4j exploitation persists years after disclosure.

05



Evolving Developer Ecosystem

Go surpassed other languages as the top API client language, capturing ~12% of automated API traffic.